TEXAS TECH UNIVERSITY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
CAPSTONE PROJECT LABORATORY

# Off-Grid Vehicle Tracker: Device User Guide

*Every Screen and Every Button, from Power On to Power Off*

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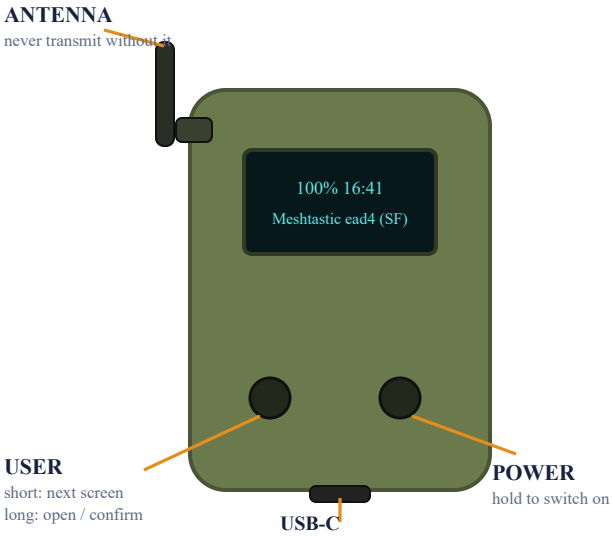
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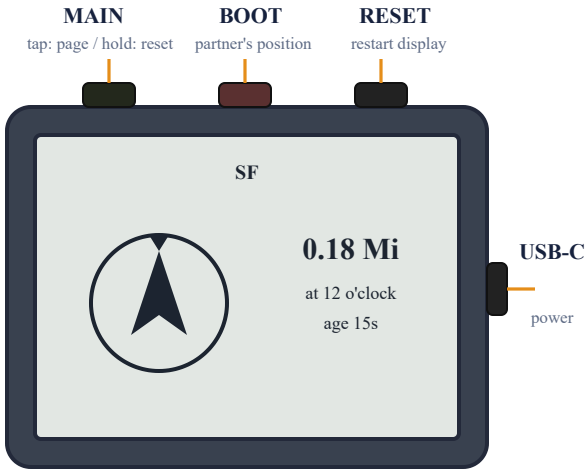

Companion document: the Technical Manual holds the engineering detail, how the system works, and every recovery procedure. This guide is only about operating the devices.

# 1 The Devices and Their Buttons

The **radio** (green box, antenna) does the talking. The **display** (dashboard screen) shows where your partner is. Four controls run everything.




**Radio (Heltec V4).** Antenna left, two buttons at the bottom.




**Display (Waveshare).** Three buttons on top, USB-C on the right.

| Control | Short press / tap | Long press / hold |
|---|---|---|
| Radio USER | Next screen or menu line | Open action menu; confirm selection |
| Radio POWER | | Switch the radio on |
| Display MAIN | Next page | Reset tracking; on Messages page, clear history |
| Display BOOT | Request partner's position now | |
| Display RESET | Restart the display (rarely needed) | |

**THE ONE RULE.** Never run the radio's battery flat. A dead battery can erase its settings, and the radios stop understanding each other until a laptop fixes it.

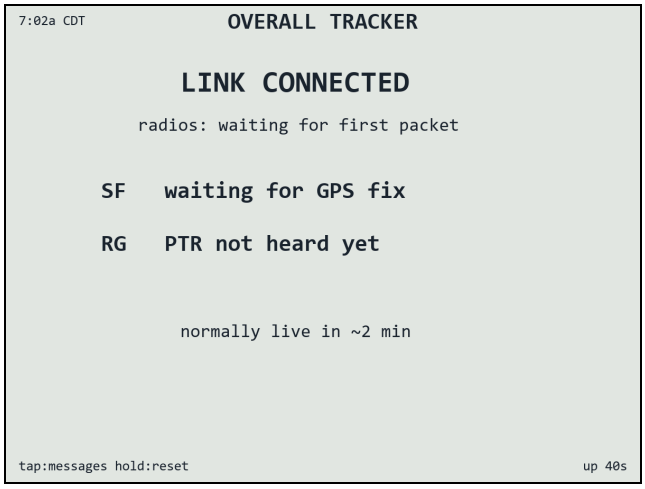
## 2 Powering On




**1. Hold the radio's power button** until the screen lights. This home screen is normal. Indoors, "0 online" and "No Sats" are not faults: satellites need sky, and your partner may be off.

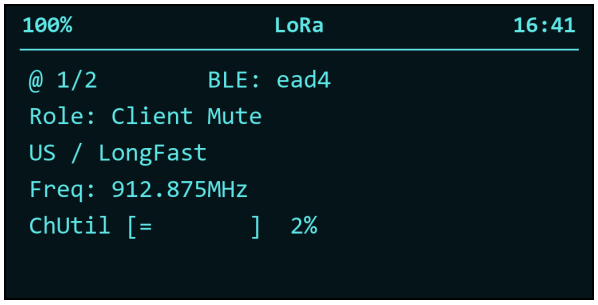



**2. Power the display at the same time**, like an ignition key. The radio accepts the display only during its first minute, so always switch them on together. Missed it? Power both off and on again.




**3. Watch the page fill in.** "waiting for GPS fix" clears under open sky; "PTR not heard yet" clears when your partner's radio is first heard. Normally live within two minutes.

## 3 The Two-Second Pre-Flight Check

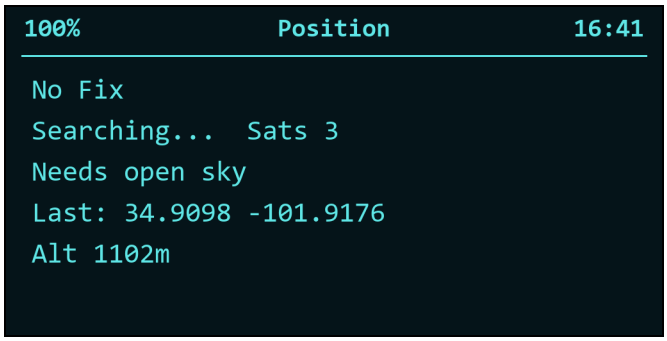



Before every trip, short-press to this **LoRa screen** on both radios and read one number: **Freq: 912.875 MHz**. Both right: everything will work. Either one different: that radio lost its settings and needs the Technical Manual's recovery before the trip.

While you are here, the rest of the screen should read Role: Client Mute and US/LongFast, exactly as shown.

## 4 The Radio's Screens

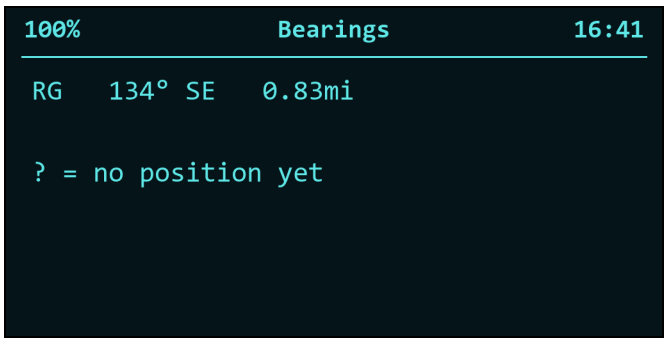
Short presses walk a loop of status screens.



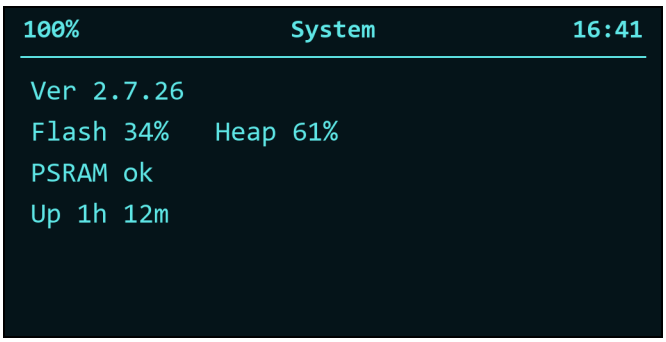

**Position.** Your own GPS. "No Fix" just means indoors.



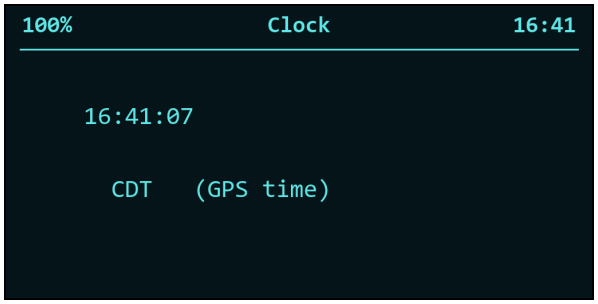

**Last Heard.** Who was heard, how long ago. The honesty meter.




**Bearings.** Direction to other radios; "?" means no position yet.



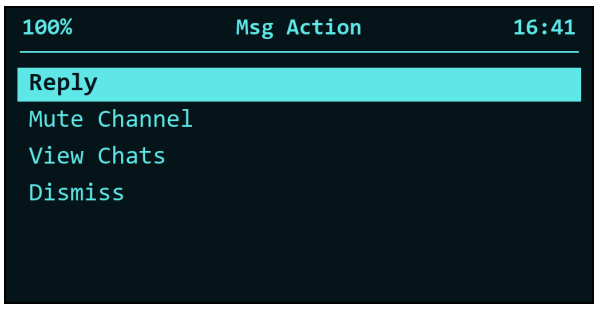

**System.** Firmware version. Needed only when asked.




**Clock.** Set by the GPS satellites; nothing to adjust. A correct clock proves the GPS has spoken to this radio.

## 5 Sending a Message

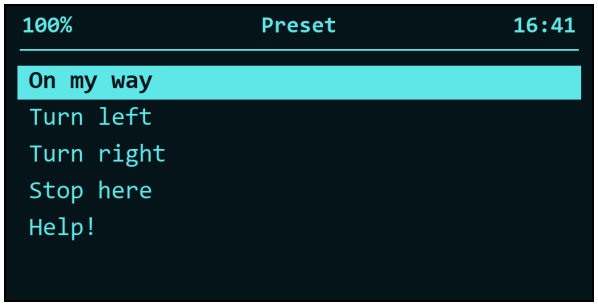
Ready-made messages, so nobody types while driving. Short press moves, long press chooses.



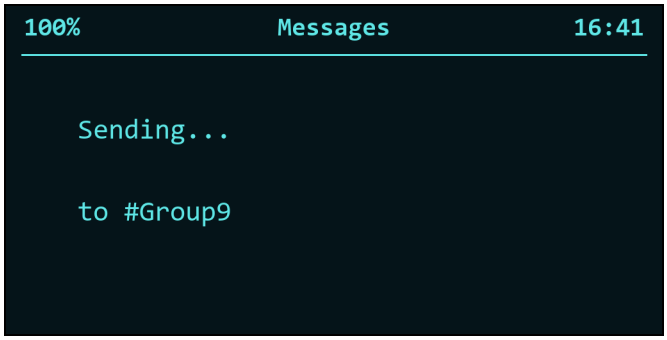

**1.** Long-press on the chat screen; choose **Reply**.




**2.** Destination: **#Group9**, the channel both vehicles hear. Skip Free Text.



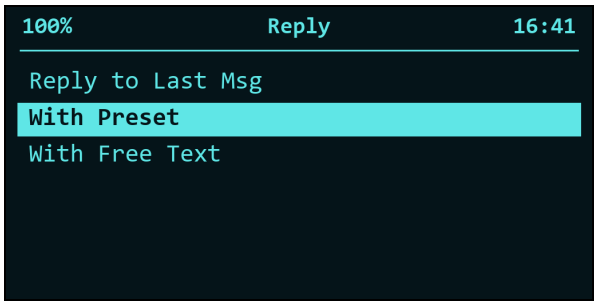

**3.** Scroll the 21 presets (Help!, Send position, On my way, Turn left, Stop...). Long-press to send.




4. **Sending.** Shown as the message goes out.



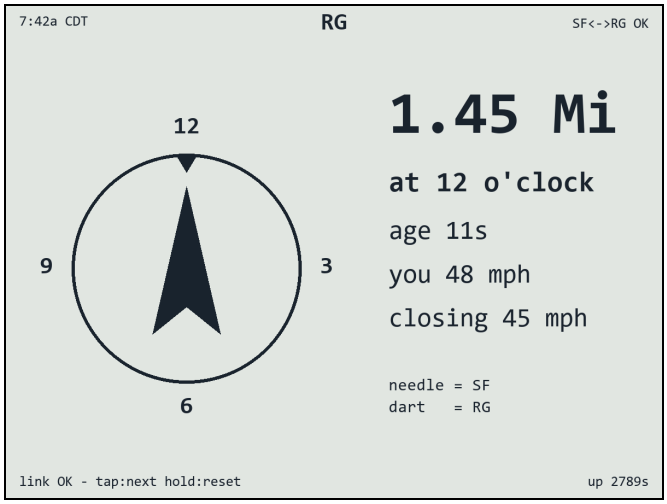

5. **Delivered.** It joins the thread and pops up on the partner's display. An X means not yet confirmed.



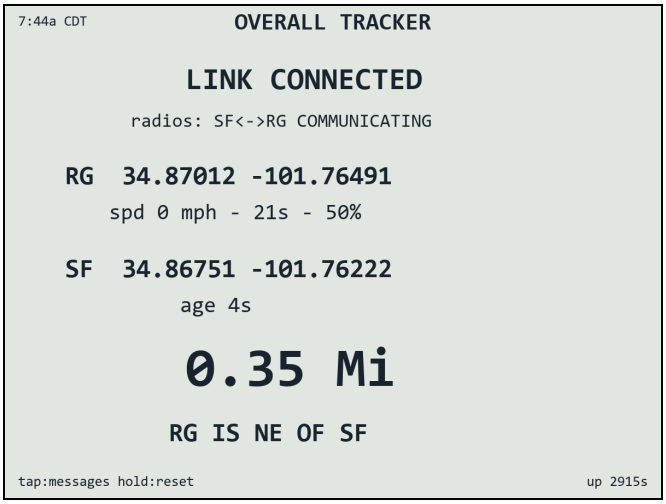

**Replying:** long-press, **Reply to Last Msg, With Preset**. Every question in the list has its answer in the list.

## 6 The Display While You Drive

Hands off: it updates itself twice a second. Tap cycles three pages.




**Compass.** The small marker at the top of the ring is your hood. The dart is your partner: steer until the dart sits under the marker and the partner is dead ahead. The ring is a clock face: 12 is always your hood, 3 your passenger side, 6 behind you, 9 your driver side, so "at 2 o'clock" on the screen and in a message mean the same thing.




**Status.** Both positions, data ages, partner battery, distance, direction in words.




**Messages.** Received texts, newest first. Hold here to clear the history.

**Partner's position now:** press BOOT. The age drops to seconds.

Proven on the road: the screen read 700 ft at 10 o'clock, the driver turned, and the partner's car appeared in the windshield. Trust the clock directions.

## 7 More Things You Can Do

Every recipe below is the same two moves: short press to reach the line, long press to choose it. All of these are safe.

| I want to... | On which device | Do this |
|---|---|---|
| Broadcast my position right now | Radio | Long press on the home screen, choose Send Position |
| Dim the screen to save battery | Radio | Long press on home, choose Sleep Screen; any press wakes it |
| Read the whole conversation | Radio | Long press on the chat screen, choose View Chats |
| Silence message pop-ups for a while | Radio | Long press on chat, choose Mute Channel; repeat to unmute |
| See when the partner was last heard | Radio | Short-press to the Last Heard screen |
| Check battery and firmware version | Radio | Short-press to the System screen |
| Switch the clock to 24-hour | Radio | Long press on the clock screen, choose Time Format |
| Fix the timezone | Radio | Long press on the clock screen, choose Timezone |
| Flip between display pages | Display | Tap the main button: compass, status, messages |
| Erase old messages | Display | On the Messages page, hold the main button |
| Start tracking fresh | Display | On any other page, hold the main button; data rebuilds in seconds |
| Get the partner's position now | Display | Press BOOT; the age drops to seconds |
| Restart a confused display | Display | Press RESET; it reconnects by itself |

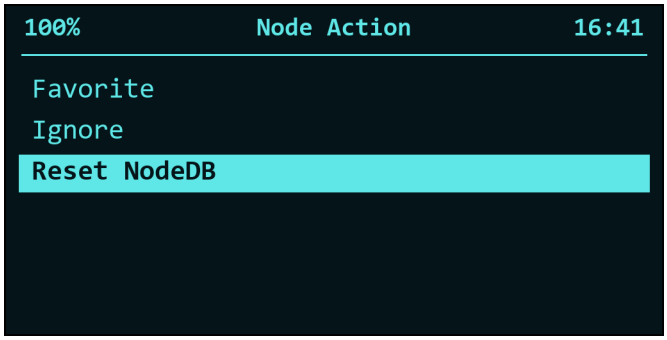
## 8 How Far It Reaches

Measured on real roads in October 2026, with the antennas inside the cars. The honest rule from all our testing: **distance is not what breaks the link, blocked line of sight is.**

| Where you are | What to expect |
|---|---|
| In town, buildings around | Reliable to about half a mile. Buildings eat radio signal. |
| Open road, flat country | Fully reliable to 3.26 miles in our test. The screens went stale by 3.75 miles. |
| Canyon or hill country | The STALE banner comes and goes at any distance, even close together, whenever rock sits between the cars. In our canyon test it appeared inside 1.3 miles, yet updates still slipped through past 4 miles when the land opened up. |
| Whenever the data is stale | The display shows DATA STALE and the age counts up. Nothing is broken, and no radio can beat rock. Keep driving toward the last known direction; the moment the two cars can "see" each other again it heals by itself, no buttons needed. |

**WANT MORE RANGE?** Mounting the antenna on the roof instead of inside the car can roughly double the distance. Ask the team before changing hardware.

## 9 Menus to Leave Alone



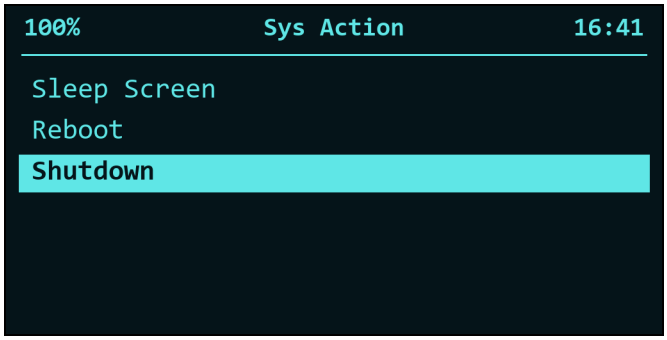

**Troubleshooting only: Node Action.** Reset NodeDB wipes known radios. Use only when the Technical Manual says so.



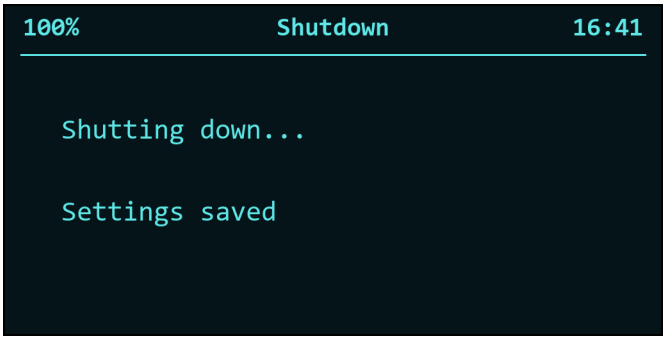

**Never change: LoRa Actions.** See the warning below.

**DO NOT TOUCH LORA ACTIONS.** Device Role, Radio Preset, Frequency Slot and LoRa Region are the exact settings that let the two radios hear each other. Changing any of them silences the pair with no error shown. Same for Bluetooth Toggle under System Action: switching it off strands your display.

## 10 Powering Off and Charging



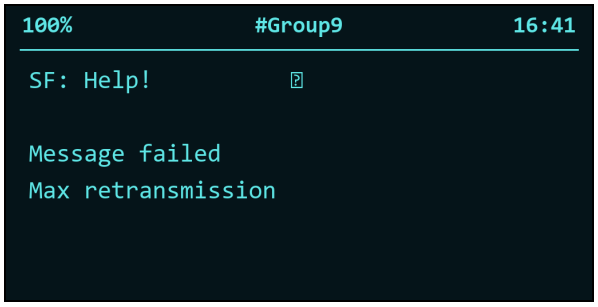

**1.** Long-press on home, open **System Action**, scroll to Reboot/Shutdown.




**2.** Choose Shutdown. A clean shutdown always preserves the settings.

Charge over USB-C after every trip. Switched off, the radio holds charge for months. Never store it empty.

## 11 When Something Looks Wrong




**"Message failed"**: nobody confirmed it; partner off, out of range, or booting. Send again once Last Heard shows them recently.

| What you see | Meaning | Do |
|---|---|---|
| 0 online, No Sats indoors | Normal | Nothing; go outside, partner on. |
| Frequency is not 912.875, or the radio's name is "Meshtastic_xxxx" | Settings lost (battery ran flat) | Section 12: five minutes with a phone, ninety seconds with the laptop. |
| Display stuck "SEARCHING" or "CONNECTING" | Missed the pairing window; if it persists, the radio lost its settings | Power both off, then on together. Still stuck after two minutes: Section 12. |
| DATA STALE while far apart | Out of radio reach | Normal past the ranges in Section 8; heals as you get closer. |
| Radios side by side, yet the display shows 0.2 mi or more, a distance that creeps, or a partner "moving" at road speed | GPS fixes are poor indoors or under a roof edge; a weak fix invents position and speed, and the display shows what the radios report | Nothing is broken. Take both radios where they can see the sky and wait two minutes; the distance settles to a few meters. Both parked reads "holding". |
| Anything else odd | Electronics | Power-cycle both together. |

**Seen on the bench, 24 October:** two radios on one desk indoors showed 0.21 mi apart, then one of them appeared to drive east at 50 mph for a minute before snapping back to 80 ft. The displays and the link were checked and were fine; the radios' GPS receivers, without a clear view of the sky, were reporting positions and speeds that did not exist. Judge distances only outdoors, and read the satellite count on the radio's Position screen (Section 4) if a number looks wrong.

## 12 Restoring a Radio That Forgot Its Settings

A radio that runs its battery completely flat can lose part of its configuration when it next starts. Nothing is broken and nothing needs re-flashing; the settings only have to be put back. You will know it has happened when one or more of these appear:

| Sign | What was lost |
|---|---|
| LoRa screen reads 906.875 instead of 912.875 | The Group9 channel (name and key), so the radio is on the public frequency and cannot hear the partner. |
| Home screen shows "Meshtastic_ead4" or "Meshtastic_bb5c" instead of SF or RG | The owner name. |
| Display stays on SEARCHING or CONNECTING for minutes | Bluetooth pairing mode went back to "Random PIN", which the display cannot use. |
| Partner position updates only every few minutes, or jumps on a coarse grid | Position timing and precision. |
| The preset message list is the factory one | The canned messages (Section 5). |

The losses come in batches, so always restore everything, not just the one thing you noticed. There are two ways to do it. The phone way needs no laptop and works at the roadside; the laptop way is faster and restores every setting at once.

### 12.1 With a phone, no laptop: about five minutes

**You need:** the Meshtastic app (free, Android or iOS) and the Group9 channel QR code. Every team member keeps the QR code in their phone's photos; it is also in the team's private hardware notes. The QR code contains the channel key: never post it, print it, or send it outside the team.

1. Switch on the radio by itself. Leave the display off until the end.
2. In the app, open the connection screen (Bluetooth) and tap the radio. It appears as **Meshtastic_ead4** (the SF radio) or **Meshtastic_bb5c** (the RG radio), or as SF or RG if the name survived. If the phone asks for a PIN, read the six digits from the radio's own screen and type them in.
3. **Restore the channel first.** Open the Channels screen, choose the scan-QR option, and scan the Group9 QR code. Accept it. The radio reboots. Check the radio's LoRa screen: it must now read **912.875**. This one step brings back the channel name, the key, the frequency, and full-precision positions.
4. **Bluetooth pairing mode.** Reconnect if the app dropped the radio, then open Radio configuration, Bluetooth, and set Pairing mode to **No PIN**. Save; the radio reboots again. Until this is done the display cannot pair.
5. **Name.** Radio configuration, User: long name and short name both **SF** (or both **RG**).
6. **Role.** Radio configuration, Device: Role **Client Mute**.

7. **Position timing.** Radio configuration, Position: GPS update interval **10 s**; position broadcast interval **60 s**; smart position on, minimum interval **10 s**, minimum distance **20 m**.
8. **Presets, optional.** Module configuration, Canned message: enabled, with the preset list from Section 5. Typed messages work without this step; only the preset list is missing.
9. Disconnect the phone from the radio in the app (the radio accepts one Bluetooth connection at a time). Now power the display: it should show the link within a minute.

> **In a hurry:** steps 3 and 4 are what the partner radio and the display need. Everything else can wait until you are parked.

### 12.2 With the laptop: about ninety seconds

Each radio has a saved snapshot of its complete, correct configuration: `sf-config.yaml` for the SF radio and `rg-config.yaml` for the RG radio, kept with the team's private hardware notes because they contain the channel key. Plug in the radio alone by USB, with the display unplugged so the radio's port is the only one, and run:

```
python -m meshtastic --configure sf-config.yaml
```

Use `rg-config.yaml` for the RG radio. **Run the command twice.** The first pass writes the region and Bluetooth settings, which make the radio reboot before the rest of the file is applied (on a fully reset radio the first pass restores Bluetooth, region and the presets but not the name, role, channel or position timing); the second pass, about twenty seconds later, completes the job. Then check the radio's own screens: the LoRa screen reads **912.875** and the home screen shows **SF** (or **RG**). If the laptop has more than one serial device attached, add `--port COMx` with the radio's port.

After any deliberate settings change, take a fresh snapshot so the saved file stays correct:

```
python -m meshtastic --export-config > sf-config.yaml
```

### 12.3 Afterwards

Switch both radios on outdoors. Within a minute, each radio's Last Heard screen should show the other, and the display should show the link. Then do the pre-flight check of Section 3 before driving. To make this section unnecessary: charge after every trip and never store a radio empty (Section 10).

## 13 Quick Reference

| I want to... | Do this |
|---|---|
| Start up | Antennas on, power radio and display together. |
| Pre-flight | LoRa screen on both radios reads 912.875. |
| Send a preset | Long press on chat, #Group9, pick, long press. |
| Reply fast | Long press, Reply to Last Msg, With Preset. |
| My position, now | Long press on home, Send Position. |
| Partner's position, now | BOOT on the display. |
| Shut down | Long press on home, System Action, Shutdown. |
| Fix almost anything | Power-cycle both; check 912.875. |
| Radio forgot its settings | Section 12: scan the Group9 QR, set No PIN; or `--configure` the saved snapshot. |
| Distance looks wrong while parked together | Indoor GPS; take both radios outside for two minutes before judging. |

## Appendix A  Abbreviations and Acronyms

| Term | Meaning |
|---|---|
| BLE | Bluetooth Low Energy; the display-to-radio link. "BLE: ead4" names the paired radio. |
| BOOT | Display button that requests the partner's position immediately. |
| CDT | Central Daylight Time, the display's timezone. |
| ChUtil | Channel utilization; how busy the shared channel is. Single digits is healthy. |
| Client Mute | The radios' role: talk and listen, never relay. Correct for two vehicles. |
| Flash / Heap / PSRAM | Radio memory types on the System screen. Never need action. |
| GPS | Global Positioning System; the satellites all positions come from. |
| #Group9 | The team's private, encrypted channel. |
| LoRa | Long Range; the radio technique carrying signals for miles on little power. |
| LongFast | The shared radio speed setting ("US/LongFast"). |
| ME / PTR | Your vehicle / the partner vehicle, on the display's status page. |
| MHz | Megahertz; a spot on the radio dial. Both radios must show 912.875. |
| NodeDB | The radio's list of radios it has heard. Reset only for troubleshooting. |
| OLED | The radio's small glowing screen. |
| RLCD | Reflective LCD, the dashboard screen; brightest in sunlight. |
| SF / RG | The two radios' callsigns. Your display shows your partner's. |
| USB-C | The charging and power plug. |
| Ver | Firmware version; the radios run Meshtastic 2.7.26. |

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Off-Grid Vehicle Tracker · Group 9, Texas Tech University ECE Capstone · Screens are exact recreations of the device screens, firmware Meshtastic 2.7.26. Range figures measured on road, October 2026.